Gradient noise can withhold recovery without weakening the drift toward it: a measured drift–diffusion dissociation in AdamW

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Abstract

From a single optimizer-complete state—identical parameters, first and second moments, step counter, and bias correction—a clean full-gradient replay re-solves the task in $\sim$800 steps, whereas a replay perturbed by both-channel gradient noise matched to the minibatch magnitude does not re-solve within 16,000 steps; both seeds are censored at that horizon, a $\geq 20\times$ slowdown that is a lower bound. The failure is not a loss of the drift toward the solution. At three states probed with paired common-random-number draws—the collapsed checkpoint and two points along a recovering trajectory—the conditional-mean update's projection onto the escape direction (the true-gradient pull) is left intact: it agrees to two decimals at the collapsed state (0.498 vs. 0.502, a $\sim 0.8\%$ difference) and matches exactly along the trajectory ($0.375/0.375$, $0.223/0.223$). We treat this drift match as a consistency check: it confirms that the nonlinear (Jensen) corrections through Adam's denominator are negligible at these matched states, a premise of the design rather than an independent result. What differs is transverse diffusion, which runs 4 to 7 orders of magnitude above the matched-drift case. The load-bearing observations are the divergent fates, this diffusion ratio, and that both hold in a real, curved loss landscape—a measured counterexample to the view of gradient noise as the agent that escapes bad regions [Jin et al., 2017, Zhu et al., 2019]. Methodologically, *where* noise is routed—not how much is injected—pins the realized noise power in update space. That routing pins the power at all is a corollary of the scale invariance of the Adam update [Kingma and Ba, 2015]; what we ship is its consequence for experiment design together with the measured level. Routing noise into the second moment holds the realized update-space power near a fixed multiple of the clean-update power: the flatness in the injected dial across the range we sweep and the $\approx 1\times$ multiple for v-only noise are algebra (construction checks on the update rule), whereas the $\approx 2\times$ multiple for both-channel noise is a measured level not forced by that algebra. The flatness holds because near small-gradient states the model does not feel the injected magnitude, so the trap persists even under reduced-amplitude both-channel noise. The noise floor deposited in the second moment overwrites Adam's per-coordinate normalization, making the v-only arm effectively a rescaled gradient-descent process and the arms genuinely different effective optimizers—a structural finding, since it is exactly what pins the update-space power. Matching noise power in gradient space is therefore not an adequate control, and we report realized update-space power throughout. We study one task at one learning rate over finite horizons; what we intend to export is the structure—a drift-preserving, diffusion-dominated trap and the routing-dependent power accounting that exposes it—not the particular multipliers.

1 Introduction

Gradient noise usually helps: it dislodges iterates from saddles [Jin et al., 2017] and, via anisotropic curvature/Fisher structure, biases toward flat basins [Zhu et al., 2019, Wu et al., 2018, Wen et al., 2020, Ziyin et al., 2024] near a stochastic-stability edge [Andreyev and Beneventano, 2024]. Adaptive methods inherit this: an Adam step is a noise-scaled sign step [Kingma and Ba, 2015, Balles and Hennig, 2018, Kunstner et al., 2023] whose second-moment denominator governs convergence [Reddi et al., 2018] and a per-coordinate noise floor [Tang et al., 2024, Li et al., 2022], rescaling not removing noise. We report the opposite. The state is *collapsed*—relaxed onto the conditional-marginal predictor, no longer using its selector, the late failure of grokking/continual learning [Prakash and Martin, 2026, Dohare et al., 2024, Chen et al., 2023]. Forking the optimizer-complete state $(\theta, m, v, \text{step}, \text{bias-correction})$ from bit-identical conditions (Fig. 1), clean full-gradient continuation re-solves in $\sim$800 steps; both-channel minibatch-magnitude noise does not, both seeds censored at $\geq 16{,}000$ ($\geq 20\times$, a lower bound).

Drift/diffusion dissociation (Fig. 3): by Monte Carlo over common-random-number draws, escape-direction pull is preserved across routings—$\mu_{\|} = 0.498/0.502$ at matched states (two decimals, MC errors), exact along a recovery trajectory ($0.375/0.375$, $0.223/0.223$)—while transverse diffusion runs 4–7 orders higher ($V_{\perp}$ ratio $3.9{\times}10^4/4.3{\times}10^6/5.3{\times}10^7$). The drift match is a *consistency check*: Jensen corrections through the Adam denominator are negligible at matched states, not a discovery. Noise slowing recovery is the *construction check* that noise impedes learning; the finding is intact pull

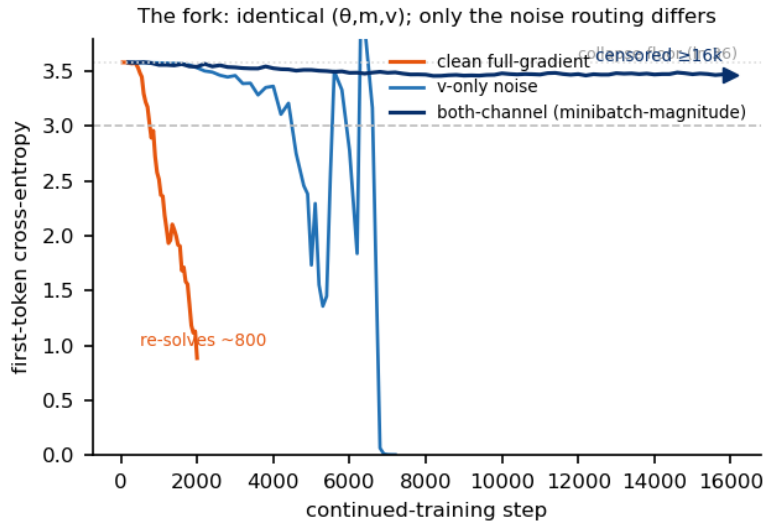

Figure 1: The fork. From one collapsed checkpoint with identical (θ, m, v) and step counter, only the noise routing differs. Clean full-gradient replay re-solves in $\sim$800 steps; both-channel minibatch-magnitude noise is censored at $\geq$16,000 steps (both seeds); v-only noise re-solves slowly. Escape times are device-sensitive (the metastable escape is float-sensitive); the relative comparison across arms uses a single harness.

coexisting with it—trapped $\|g\|$ grows $0.048 \to 0.583$, so noise buries the pull rather than reaching a drift-dead region, a counterexample to "noise=escape agent" scoped to these states.

Realized-power pinning (Fig. 2): routing pins update-space power—v-only holds $\Gamma \approx 1$, both-channel Γ flat in its dial, m-only explodes ($\Gamma \gtrsim 250$ vs clean-v)—*construction checks*, algebra of Adam's scale invariance/denominator [Kingma and Ba, 2015], whence pinning follows as a corollary. The shipped claim controls that corollary plus a measured level: gradient-space power matching is inadequate—report update-space power—as reduced-amplitude both-channel noise stays censored at $\geq 16{,}000$; both-channel $\Gamma \approx 2$ (measured 1.89 vs scalar $1+a^2 = 1.006$; $q = 0.70$, $\rho = -0.28$) is unexplained. Own-axis amplification (clean $\times 145$, v-only $\times 106$; P_T $741/166$) and the floor injected into v inflate the second moment (*construction check*); the finding: inflation overwrites Adam's per-coordinate normalization, so v-only continuation is rescaled plain gradient descent and the arms are different effective optimizers.

Frozen-fork methodology (Fig. 1): each comparison forks one optimizer-complete state, freezes escape criteria in advance, censors non-escapes at the horizon, carries Γ shadow-accounting, and checks against a hand-written AdamW reproducing the replay to 1.3×10^{-17} (float32 matching float64) [Biderman et al., 2026]. A 2×2 over initialization and routing (collapsed vs. fresh $\times$ clean vs. v-only; Fig. 4) closes the loop: the collapsed head start is eliminated, possibly inverted, under noise (onsets $800/4600/2200/8400$)—readable with equal weight as noise targeting recovered structure or both conditions relaxing to a common noise-set onset floor.

2 Setup

Each comparison contrasts optimizer runs sharing a start state, differing in one controlled respect [Biderman et al., 2026].

System and collapsed state. A small transformer trained with AdamW [Kingma and Ba, 2015] memorizes 10,000 rows $(B, z) \mapsto A$, B K-fold ambiguous, z disambiguating. Continued full-batch training past interpolation drives it onto the *input-blind marginal*: first-token CE rises to $\approx \ln 36 \approx 3.58$ nats, the measured optimum (not $\ln K$, a candidate-restricted diagnostic). This collapsed checkpoint, a degenerate attractor of continued training [Chen et al., 2023], starts all forks.

The frozen fork. From one checkpoint we fork copies sharing the full optimizer state (θ, moments (m, v), step, bias-correction), differing *only* in where injected noise routes into the moments: **clean** (none); **v-only** (second moment noised, first clean); **both-channel**; **reduced-amplitude both-channel**; **m-only** at low dose (first-moment noise over a clean second moment drives the per-coordinate denominator toward zero and diverges—a construction check, not a result); **minibatch-magnitude** (both-channel, variance matched to the measured minibatch-gradient fluctuation). A finding, not a caveat: a noise floor in v swamps the gradient in Adam's per-coordinate denominator, so v-only is a *rescaled gradient-descent process* and the arms are strictly different effective optimizers—the mechanism collapsing DP-Adam onto DP-

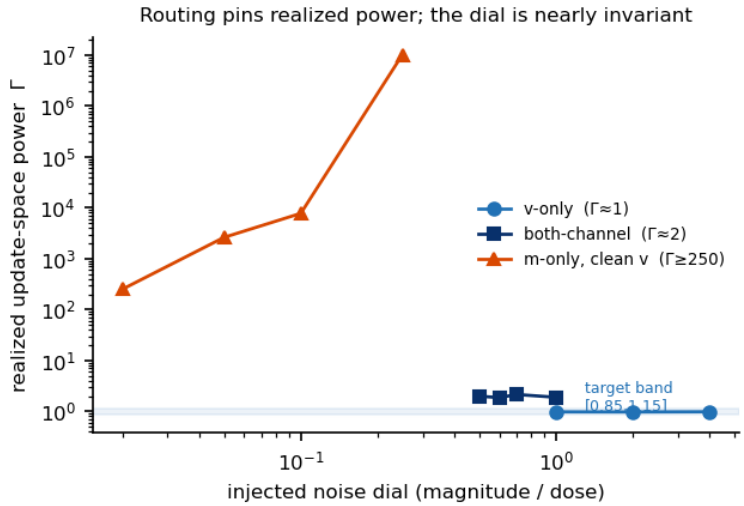

Figure 2: Routing pins realized power. Realized update-space noise power Γ (shadow-update power ratio) versus the injected dial. v-only sits at $\Gamma \approx 1$, both-channel at $\Gamma \approx 2$, and direction-only (m-only) noise at low dose already realizes $\Gamma \geq 250$ and grows. Within each routing, Γ is nearly dial-invariant; reduced-amplitude both-channel noise is also outcome-invariant (still censored).

SGD [Tang et al., 2024, Li et al., 2022]. *Where* noise enters, not how much, sets the dynamics.

Solve/onset criteria and $\geq$-horizon censoring. A fork is *solved* when first-token CE returns to and sustains the pre-collapse band; *onset* is first entry. Both criteria and the 16,000-step horizon were fixed a priori; an unsolved fork is right-censored at $\geq 16{,}000$ steps as a lower bound, never imputed. On the reference fork (Fig. 1), clean replay re-solves in ≈ 800 steps; both-channel minibatch-magnitude noise does not, both seeds censored at $\geq 16{,}000$ (final CE $3.47/3.54$ vs the 3.58 floor)—a $\geq 20\times$ slowdown, itself a lower bound. Absolute times are device-sensitive; every cross-arm comparison runs in one harness.

Realized-power (Γ) shadow accounting. A *shadow* optimizer advances its moments by the clean full-batch gradient at θ_t, giving the clean-step update u_t^{sh}; the realized update-space noise power

$$\Gamma = \frac{\sum_t \|u_t - u_t^{\text{sh}}\|^2}{\sum_t \|u_t^{\text{sh}}\|^2}$$

is a windowed median over steps 100–500. Pinning Γ almost independently of the injected dial is a corollary of Adam's scale invariance [Kingma and Ba, 2015]: rescaling amplitude leaves per-coordinate normalization unchanged, so a flat Γ-versus-dial curve is algebra, a construction check; the shipped claim controls that corollary and adds the measured level. Gradient-space matching is inadequate—the model barely feels injected magnitude near small-gradient states—so we report realized Γ. The both-channel level $\Gamma \approx 2$ doubles the ≈ 1 of an independent scalar-noise account, and both-channel noise censors at $\geq 16{,}000$ even at reduced amplitude ($\Gamma \approx 1.86$): the trap tracks pinned realized power, not the dial (Fig. 2).

Replay and precision verification. A hand-recomputed one-step AdamW update from $(\theta, m, v, \text{step})$ matches the framework to relative error 1.3×10^{-17}; the conditional-update statistics behind Fig. 3 are bit-identical in float32 and float64. Reported gaps between arms are dynamical, not numerical.

3 The fork and sufficiency

We take a single collapsed checkpoint—a run that fit then relapsed to the conditional-marginal floor [Prakash and Martin, 2026]—and *fork* it into an optimizer-complete replay: identical θ, moments m, v, step counter, and bias-correction state, differing only in gradient signal, so any divergence isolates the injected noise. Controls: our manual AdamW [Kingma and Ba, 2015] step reproduces the framework optimizer to relative 1.3×10^{-17}; float32 and float64 replays are bit-identical; the solve criterion is frozen before the fork; every non-escape is reported right-censored at the horizon.

The fork. The clean full gradient re-solves in $\sim$800 steps (Fig. 1). *Both-channel noise matched to the minibatch magnitude* (into both m and v) does not re-solve within 16,000 steps for either seed; both stay at the floor (cross-entropy

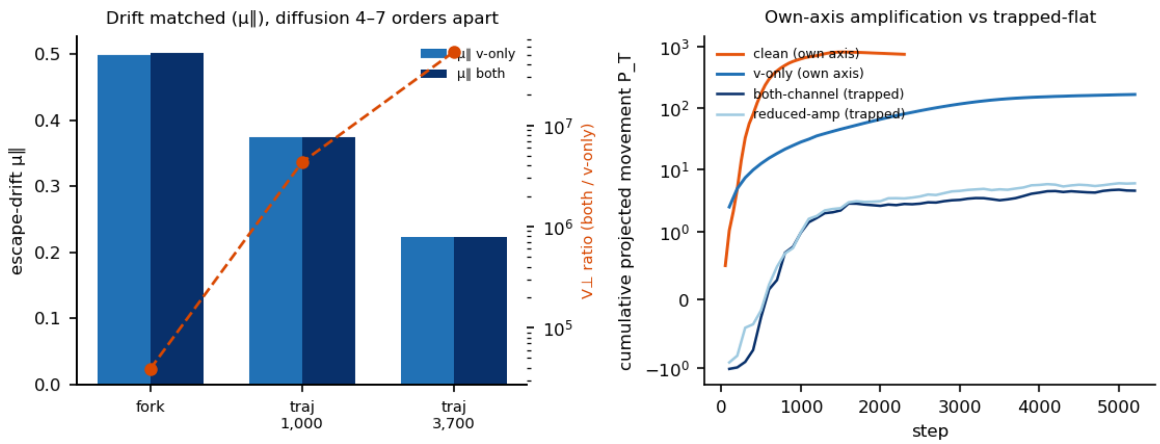

Figure 3: The dissociation. Left: the escape-direction drift $\mu_{\|}$ is matched between v-only and both-channel at the fork (to two decimals) and along the v-only trajectory (exactly), while the transverse diffusion ratio $V_{\perp}^{\text{both}}/V_{\perp}^{\text{v}}$ runs 4–7 orders of magnitude above it. Right: on each recovering arm's own escape direction, cumulative projected movement P_T grows (amplification); the censored arms stay flat.

3.47, 3.54 vs. $\ln 36 \approx 3.58$). The censored horizon is thus a lower bound: a $\geq 20\times$ slowdown. Escape times are device-sensitive. The v-only arm (same panel) is itself a finding: a persistent noise floor in v saturates Adam's per-coordinate normalization, so it evolves as rescaled plain-gradient descent.

Sufficiency, and of what. Minibatch-magnitude noise is thus *sufficient* to hold the model at the floor. At fixed realized update-space power, two covariance structures—isotropic, and the empirical minibatch covariance—both censor at $\geq$ 16,000 steps: scaled stochastic *power* alone suffices, the anisotropic structure unnecessary and only tightening the trap. Reducing the both-channel amplitude to $c = 0.6$ (realized $\Gamma \approx 1.86$) still censors at $\geq 16{,}000$ steps (Fig. 2). Flat $\Gamma(c)$ is a *construction check*—post-hoc algebra from Adam's scale invariance [Kingma and Ba, 2015]. The shipped claim is the methodological corollary (control the realized power Γ, not the injected amplitude) plus the measured anomaly that Γ pins near 2, above the scalar prediction.

This aligns with stochastic collapse [Chen et al., 2023] and dynamical-stability accounts where noise magnitude, not presence, governs retained minima [Wu et al., 2018, Andreyev and Beneventano, 2024]; the matched-power fork adds a clean separation of magnitude from covariance [Zhu et al., 2019] at a fixed collapsed state in a real curved landscape.

4 Channels and pinning

The fork of Fig. 1 turns on *where* a gradient perturbation enters AdamW. With $u_t = \hat{m}_t/(\sqrt{\hat{v}_t} + \epsilon)$ and $\tilde{g}_t = g_t + \xi_t$, the same ξ_t routes into three arms sharing frozen (θ, m, v): **v-only** (clean $\hat{m}$, noisy $\hat{v}$); **both-channel** (both moments from $\tilde{g}_t$, what a minibatch does; unit dial calibrated to per-step minibatch gradient magnitude); **m-only** (noisy $\hat{m}$, clean $\hat{v}$).

v-only is a different optimizer (finding, not caveat). A second-moment noise floor inflates $\hat{v}_t$ (v-inflation, a construction check). Near a collapsed checkpoint's small-gradient states $\sqrt{\hat{v}_t}$ is set by the injected variance, so Adam's per-coordinate normalization washes out and u_t reduces to a rescaled clean first moment—plain gradient descent. So v-only and both-channel are different effective optimizers, not one at two noise levels: the v floor overwrites Adam's normalization, as when noised-gradient Adam collapses onto DP-SGD unless corrected [Tang et al., 2024, Li et al., 2022], consistent with the sign/variance reading [Balles and Hennig, 2018, Kunstner et al., 2023].

Routing pins realized update-space power. What the model feels is the realized power ratio Γ (perturbed-minus-shadow update power over shadow power, median over steps 100–500; u^{sh} the clean update), not the injected magnitude. Across the dial Γ is pinned by routing (Fig. 2): v-only $s{=}1, 2, 4 \to \Gamma \approx 1$; both-channel $c{=}0.5, 0.6, 0.7, 1.0 \to 1.99, 1.86, 2.15, 1.89\,(\approx 2)$; m-only $s{=}0.02, 0.05, 0.1, 0.25 \to 251, 2618, 7773, 1{\times}10^7\,(\geq 250)$. Each is a construction check: *v-only* $\Gamma_v \approx 1$, the floor absorbed by the denominator it inflates; *both-channel* flat in c by algebra, as Adam's scale invariance [Kingma and Ba, 2015] makes $\hat{m}/\sqrt{\hat{v}}$ independent of rescaling $\tilde{g}_t$; *m-only* explodes, noise in $\hat{m}$ with clean $\sqrt{\hat{v}}$ small inflating the step [Reddi et al., 2018].

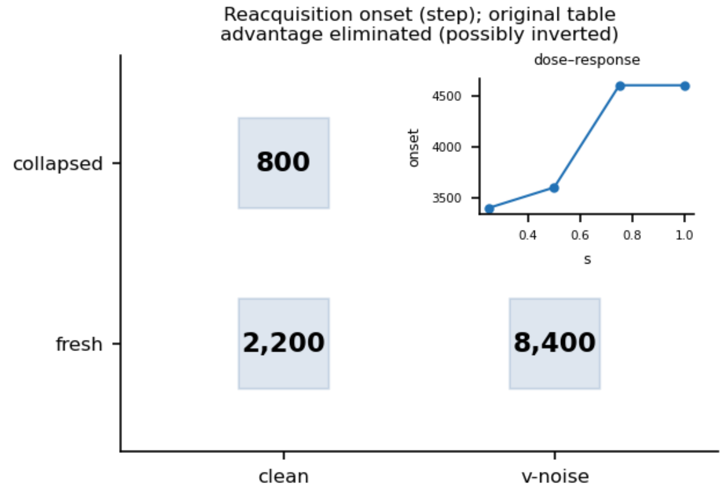

Figure 4: Reacquisition 2×2**.** Onset step by starting state and routing (original table). Fresh–clean learns at this rate, so the rate is not the obstruction; under v-only noise the trained-state advantage is eliminated (possibly inverted). Inset: v-only onset saturates in the injected scale.

Outcome tracks Γ, not the dial. Reduced-amplitude both-channel noise ($c = 0.6$, $\Gamma \approx 1.86$) censors re-solving for $\geq 16{,}000$ steps just as the minibatch-magnitude setting does (Fig. 1); that added noise impedes learning is generic (a construction check), the content being that fate follows Γ, not the dial. So gradient-space power matching is not an adequate control: interventions matched on injected magnitude differ by orders of magnitude in realized power—report Γ or shadow-update power, not the dial.

Derivable core and residue. The pinning core—Γ nearly dial-independent within each routing—is derivable post hoc from Adam's scale invariance [Kingma and Ba, 2015] and not claimed as a discovery; the shipped claim is that methodological corollary plus the one quantity the invariance does not fix: the both-channel *level*. A scalar-noise account predicts $\Gamma = 1 + a^2 = 1.006$ at floor $a = 0.079$, but the measured level is 1.89—a $\sim 90\%$ excess. It is a turning, not an inflation: the realized update is smaller ($q = \|u_{\text{both}}\|/\|u_{\text{clean}}\| = 0.70$) yet anti-aligned ($\rho = \cos(u_{\text{both}}, u_{\text{clean}}) = -0.28$), and $\Gamma = 1 + q^2 - 2q\rho = 1.894$ recovers it. We carry $\Gamma \approx 2$ as a measurement; a sparsity account (clean update on $\sim 0.3\%$ of coordinates, noise dominating the near-zero remainder where normalization amplifies it) is in the appendix. How this power denies recovery yet leaves the escape-direction pull intact is next (Fig. 3).

5 The mechanism measurement

Two routings, two effective optimizers. We bracket the fork with a v-only routing (clean m, noisy v) and a both-channel routing. v-only realizes update-space power $\Gamma \approx 1$—the level its second-moment algebra predicts, a construction check (Fig. 2)—while both-channel realizes $\Gamma \approx 2$. That $\Gamma_v \approx 1$ pinning is derivable post-hoc from Adam's scale invariance [Kingma and Ba, 2015], so we ship not the pinning but its corollary—gradient-space power matching is not an adequate control, so realized update-space power must be reported—together with the both-channel level (measured $\Gamma = 1.89$ vs. a scalar-noise prediction $1 + a^2 = 1.006$; $q = 0.70$, $\rho = -0.28$), an anomaly that account does not predict. Its *consequence* we report as a finding: a noise floor in the second-moment accumulator swamps Adam's per-coordinate denominator and removes its normalization, so the v-only arm evolves as a *rescaled plain-gradient-descent* process—the mechanism by which DP-Adam collapses onto DP-SGD once bias-corrected noise reaches the denominator [Tang et al., 2024, Li et al., 2022], turning on Adam's identity as a normalized sign/variance step [Kingma and Ba, 2015, Balles and Hennig, 2018, Kunstner et al., 2023, Reddi et al., 2018]. Not a caveat: the two arms are genuinely different effective optimizers, and the dissociation below concerns routings and their realized diffusion, not one fixed optimizer.

Drift matched, diffusion four to seven orders apart. We measure the conditional one-step update at the fork (K=512 paired draws, common random numbers; $\mathbb{E}[u]$ Monte-Carlo averaged, not by construction; float32/float64 bit-identical; Table 1). Along $e = u_{\text{clean}}/\|u_{\text{clean}}\|$ the drift $\mu_{\|} = \langle \mathbb{E}[u], e \rangle$ agrees to two decimals (0.498 vs. 0.502); the resolved gap $\Delta\mu_{\|} = +0.0038$ (bootstrap 95% CI $[0.0015, 0.0062]$, $\approx$0.8%) is small and, if anything, marginally favors the trapped arm. Transverse diffusion differs $\times$3.9e4 (0.021 vs. 817). This matched drift is a *consistency check*, not a discovery: it verifies a premise—that nonlinear (Jensen) corrections through the Adam denominator are negligible at matched states.

Table 1: Conditional one-step update at the fork (K=512 paired draws, common random numbers; $\mathbb{E}[u]$ Monte-Carlo averaged, not by construction; float32/float64 bit-identical). $e = u_{\text{clean}}/\|u_{\text{clean}}\|$.

	v-only noise	both-channel noise
escape drift $\mu_{\|}$	$0.498 \pm 1.1{\times}10^{-4}$	$0.502 \pm 1.2{\times}10^{-3}$
$\cos(\mathbb{E}[u], u_{\text{clean}})$	0.933	0.366
transverse diffusion $V_{\perp}$	0.021	817 (×3.9e4)
along-axis $\text{SNR}_{\|}$	193	18.5

The load-bearing content is the fates (re-solve vs. censored $\geq$16,000 steps), the diffusion gap, and that both hold in a real curved landscape. Along the v-only arm's own trajectory (K=256) the drifts coincide at reported precision (0.375/0.375 at step 1,000; 0.223/0.223 at 3,700) while the transverse-diffusion ratio (both/v-only) climbs 4.3e6→5.3e7 toward onset. Across all three measured states the pull toward re-solution is preserved and diffusion runs four to seven orders above it. This is the counterexample to "noise=escape agent"—where injected noise usually carries a trajectory off a plateau [Jin et al., 2017, Zhu et al., 2019] or selects minima [Wu et al., 2018, Ziyin et al., 2024, Andreyev and Beneventano, 2024, Wen et al., 2020], here the noise that denies recovery leaves the pull intact and drowns it transversely [Chen et al., 2023]—and the fate tracks realized diffusion, not amplitude: the reduced-amplitude both-channel arm ($\Gamma \approx 1.86$) stays censored $\geq$16,000 steps.

Burial, and its mirror image in recovery. At the censored both-channel states the full-batch gradient norm is nonzero and *grows* across the trap ($\|g\|$ 0.048 → 0.583 over 0.5k–16k) with a small consistently-signed escape-direction component: an intact pull that transverse diffusion buries, not a drift-dead region to which noise relocated the state—clean replay from the same weights re-solves in ≈800 steps. This separates the phenomenon from loss of plasticity, where the useful gradient itself decays [Dohare et al., 2024], and from late-stage collapse read as an endpoint [Prakash and Martin, 2026]. Recovery is the mirror image, amplification: each recovering arm on its *own* escape direction shows systematic pre-onset growth of $s_t = \langle -g, e_{\text{own}} \rangle$—cumulative projected movement P_T of 741 for clean (×145) and 166 for v-only (×106), each direction internally stable (pairwise $\cos = 0.97$). The directions differ *across* arms: v-only recovers nearly orthogonal to clean, its P_T only 9.9 on clean's axis vs. 166 on its own, so a single held-out direction does not transfer and the amplification is arm-specific. Together: recovery amplifies an intact escape-direction pull along the arm's own axis, and trapping leaves that same pull intact while transverse diffusion prevents its amplification—the fate set by realized diffusion, not drift; the object of study is the trajectory, not the endpoint [Biderman et al., 2026].

6 The reacquisition 2×2

To separate whether the noise acts on *what collapse stored* or merely on *learning in general*, we cross initialization (fresh random init vs. the collapsed checkpoint [Chen et al., 2023]) with routing (clean vs. v-only noise at s=1.0), one seed per cell at the fixed η=0.0125, reporting the onset step (Fig. 4). We use v-only here—not both-channel, which is censored at $\geq$16,000 steps—so every cell reaches onset. Routing noise into v alone pins realized power at $\Gamma \approx 1$ by construction (a scale-invariance identity; Fig. 2); its reportable consequence is structural: the second-moment noise floor overrides Adam's per-coordinate normalization, so the v-only arm is a rescaled plain-gradient-descent process. Clean and v-only are thus *different effective optimizers*, not one optimizer at two noise levels—the destruction of Adam's normalization is the mechanism the grid probes, not a confound.

The grid: collapsed = 800 (clean) / 4,600 (v-only); fresh = 2,200 (clean) / 8,400 (v-only). Fresh+clean reaches onset at 2,200, so the task is acquired from scratch at this rate and no elevated onset is a rate artifact. That v-only noise slows the fresh run (2,200→8,400) is a *construction check*—generic noise impedes learning—not a finding; the signal is how the rows move *relative* to one another. In the clean column collapse buys a head start (800 vs. 2,200, 2.75×); in the noise column that relative advantage all but disappears (4,600 vs. 8,400, 1.83×), the collapsed cell absorbing the larger multiplicative penalty (5.75× vs. 3.8×). We record the registered claim: the collapse advantage is *eliminated (possibly inverted)*. The grid replicates on the original table (promoted from provisional); an earlier sibling table read 4,000 in the fresh+v-only cell, a table artifact we do not build on. On the original table the fresh+v-only onset is 8,400 > 4,600, so the head start survives in absolute terms and we make *no standalone inversion claim*.

We read the noise cells two ways, at equal weight. Under *targeted erasure*, the noise degrades precisely the conditional structure that made the collapsed run fast, so reacquisition proceeds from a partly-erased prior [Dohare et al., 2024]; the collapsed cell's 5.75× slowdown is its signature. Under a *common onset floor*, v-only noise imposes an escape-time cost largely independent of starting structure, lifting both rows toward a shared noise-set floor; the near-parity of the two penalties and the compression of the row ratio ($2.75\times \rightarrow 1.83\times$) are its signature. Both are consistent with an eliminated advantage and neither requires inversion. The elevated collapsed onset saturates in noise scale (3,400/3,600/4,600/4,600 at s=0.25/0.5/0.75/1.0; Fig. 4 inset), mirroring the dial-invariance of realized power.

7 Related work

We treat late-training regime change as an object of dynamical study [Biderman et al., 2026]; our collapsed checkpoint resembles the destination of late-stage generalization collapse in grokking [Prakash and Martin, 2026], but the question is what a chosen noise process does once the network is there.

Noise as escape agent and regularizer. A large literature reads gradient noise as the agent that leaves bad regions and shapes solutions: curvature-aligned anisotropic SGD noise [Zhu et al., 2019], its Fisher structure [Wen et al., 2020], provable saddle escape under injected perturbations [Jin et al., 2017], collapse onto low-rank invariant sets [Chen et al., 2023]. We report a measured counterexample (Figs. 1, 3): clean replay re-solves in $\sim$800 steps while the both-channel replay is censored at $\geq$16,000; at the three probed states the escape-direction drift is matched across routings (two decimals at the fork, exact along-trajectory) while transverse diffusion runs 4–7 orders higher for the censored routing. The matched drift is a consistency check (Jensen corrections through the Adam denominator negligible at matched states), not a claim; the load-bearing facts are the opposite fates, the diffusion ratio, and that both hold in a real curved landscape. Scoped to these states, noise denies recovery without weakening the pull toward it.

Which minima noise selects. Dynamical-stability analyses explain which minima SGD keeps [Wu et al., 2018], symmetry-induced equilibria describe where its noise balances [Ziyin et al., 2024], and the edge of stochastic stability sets an operating point [Andreyev and Beneventano, 2024]. Our trapped state is not a destabilized minimum—the true gradient is nonzero and grows across the horizon ($\|g\|$ $0.048 \rightarrow 0.583$) with a small consistently-signed escape-direction pull—so the pull is intact but buried, and the operating point is set by *where* noise enters Adam's moments, not by a curvature balance.

Adam's sign/variance channel and the SGD–Adam gap. Adam's second-moment normalization behaves as a sign/variance-adaptive step [Kingma and Ba, 2015, Balles and Hennig, 2018], its convergence depends on how that average is formed [Reddi et al., 2018], and the SGD–Adam gap is driven more by sign structure than noise [Kunstner et al., 2023]. Our pinning acts on this channel: a noise floor in the second moment removes per-coordinate normalization, so the v-only arm is a rescaled gradient-descent process—stated as a finding, the arms being structurally different effective optimizers. Flat Γ across the both-channel dial follows post-hoc from Kingma–Ba scale invariance (an algebraic construction check); the shipped contributions are the reporting corollary below and the measured level anomaly ($\Gamma = 1.89$ vs. scalar 1.006), traced to Adam amplifying noise on the $\sim$99.7% near-zero-gradient coordinates.

Differentially private Adam: a reporting consequence. DP-Adam is DP-SGD unless bias correction is applied [Tang et al., 2024], and DP fine-tuning sets its budget in injected gradient magnitude [Li et al., 2022]. Pinning implies that near small-gradient states an injected-magnitude budget does not determine realized update-space power under Adam (Fig. 2), so a gradient-space power match is not an adequate control: realized update-space power should be reported.

Plasticity: fresh versus aged. Loss of plasticity attributes failure-to-learn in aged networks to a capacity deficit [Dohare et al., 2024]. Our fresh-vs-aged 2×2 (Fig. 4) isolates a different axis: the aged (collapsed) start is not learning-disabled (both clean cells reach onset quickly, 800 vs. 2,200), so the obstruction under v-only noise is the routed noise, not aging. We read the noise cells two ways at equal weight, targeted erasure and a common onset floor; the advantage is eliminated (possibly inverted), with no standalone inversion claim. The variable governing the fate is not the injected magnitude but where the optimizer routes it [Biderman et al., 2026].

8 Limitations and corrections

Finite horizons. Every non-escape is right-censored, not a claim escape cannot occur: both-channel arms did not re-solve within 16,000 steps (both seeds), so $\geq$16,000 bounds escape time below and the $\geq 20\times$ gap from the $\sim$800-step clean re-solve is likewise a lower bound (Fig. 1). Absolute times are device/precision-sensitive (slower on MPS, near a float boundary); we quote steps under one harness.

One system, one rate. One task, architecture, and rate (2×2 reacquisition at $\eta = 0.0125$, single seed per cell and dose; only fork censoring replicated, $n = 2$). We read the noise column as *advantage eliminated (possibly inverted)*, weighting equally its two readings—targeted erasure of the trained head start, and a common onset floor both start states reach—without headlining inversion (Fig. 4). What exports is *structure*—routing pins realized update-space power, dissociating drift from diffusion, denying reacquisition at intact drift—not the multipliers ($\times$39,000 diffusion ratio, 4–7 orders along the trajectory, $\geq 20\times$ slowdown, Γ levels). Reduced-amplitude both-channel noise ($\Gamma \approx 1.86$) is still censored: the trap tracks pinned felt-noise, not the injected dial (Fig. 2).

What the drift match buys. Equality of the escape drift $\mu_{\|}$ across arms is a consistency check, not a discovery: 0.498 vs 0.502 at the fork (resolved, $\sim 0.8\%$ of drift), exact along the v-only trajectory ($0.375/0.375$ at step $1{,}000$, $0.223/0.223$ at step $3{,}700$)—confirming only that Jensen corrections through the Adam denominator are negligible *at matched states*, a verified premise, not a result. Load-bearing are the fates, the transverse-diffusion ratio, and that both hold in a real curved landscape (Fig. 3); our strongest phrasing (noise does not weaken the pull) is scoped to the three probed states.

Different effective optimizers, as a finding. The v-only and both-channel arms are, by construction, different effective optimizers—mechanism, not confound. A noise floor in the second moment v inflates it toward a near-constant, so the v-only update reduces to a rescaled clean-gradient (momentum) step and Adam's per-coordinate normalization is overwritten—the route by which noise in v turns DP-Adam into rescaled DP-SGD absent a bias correction [Tang et al., 2024], seen in private fine-tuning [Li et al., 2022] and consistent with the SGD/Adam gap living in normalization, not noise magnitude [Kunstner et al., 2023]. Hence v-only realizes $\Gamma \approx 1$. The second-moment inflation, flat $\Gamma(c)$, $\Gamma_v \approx 1$, and m-only denominator blow-up are construction checks following from Adam's scale invariance [Kingma and Ba, 2015]. Pinning is a corollary of that invariance; we ship the method corollary—gradient-space power matching is not an adequate control, so realized update-space power must be reported—plus the measured $\Gamma \approx 2$ level, which is empirical, not algebraic.

Corrections ledger. App. A logs withdrawn claims: escape impossibility (now $\geq$-horizon censoring), erasure-to-scratch, second-moment-channel-alone, a crawl/round-trip overcount, "ignition" and "corridor" demoted by their registered tests, and one horizon deviation (T: $3{,}500 \rightarrow 2{,}000$).

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A Run ledger

Every arm reported in the paper draws on one collapsed checkpoint (the fork state) or a fresh initialization, run under a single AdamW harness with the same horizon. Non-escapes are censored: an arm that has not re-solved by the horizon is recorded as "$\geq$-horizon," a lower bound on its reacquisition time, not as a terminal outcome. Escape steps are the onset step at which the model leaves the $\ln 36$ floor. Table 2 is the complete accounting behind Figs. 1 and 4; the dose sweep is broken out in App. E. That noise slows learning at all is a construction check, not the result; the result is that the slowdown is set by *where* the noise is routed [Biderman et al., 2026].

Routing	Start	Seeds	Outcome	Note / deviation
Clean (full-grad. replay)	collapsed fork	— (noise-free)	escape $\sim$800	CPU decomposition; MPS escape slower (device-sensitive; see App. C)
Both-channel (minibatch-magnitude)	collapsed fork	2 (both censored)	$\geq$16,000; final CE 3.47 / 3.54	floor $\approx$ 3.58 $=$ $\ln 36$; the $\geq$20$\times$ slowdown is a lower bound
Reduced-amplitude both-channel (c=0.6, $\Gamma\approx$1.86)	collapsed fork	1	$\geq$16,000 (censored)	outcome-invariance: trap tracks pinned felt-noise, not injected amplitude
Minibatch-magnitude	collapsed fork	1	displayed reference (Fig. 1)	the magnitude the both-channel arm is matched to
v-only noise (s=1.0)	collapsed	1	escape 4,600	dose row, App. E
Clean	fresh init	— (noise-free)	escape 2,200	the rate (η=0.0125) is not the obstruction
v-only noise (s=1.0)	fresh init	1	escape 8,400 (original table)	sibling-table 4,000 was a table artifact, excluded (App. C)

Table 2: Run ledger. Horizon $=$ 16,000 steps throughout; 2×2 onsets are at $\eta = 0.0125$, single seed per cell. Registered deviation applied uniformly: the stuck-state detection threshold T was reduced 3,500 $\to$ 2,000 mid-program (logged in App. C).

B The realized-power pinning table

Table 3 is the full version of Fig. 2: realized update-space noise power $\Gamma = \sum\|u - u_{\text{sh}}\|^2 / \sum\|u_{\text{sh}}\|^2$ (shadow-update power ratio), windowed-median over steps 100–500, against the injected dial. The core is derivable post-hoc from Adam's scale invariance [Kingma and Ba, 2015]: the flatness of Γ across the both-channel dial c is algebra, and $\Gamma_v \approx 1$ for v-only is the same v-inflation identity—injecting into an already noise-dominated denominator adds no update-space power. Both are construction checks. The m-only column exploding is also a construction check (the clean-v denominator: momentum noise divided by a near-zero, clean second moment blows up per coordinate), in line with the AMSGrad denominator analysis [Reddi et al., 2018]. The shipped claim is the control corollary and the measured $\Gamma \approx 2$ level (App. D), not the invariance.

Routing	Injected dial	Realized Γ (median, steps 100–500)
v-only (clean m, noisy v)	$s = 1.0$ / 2 / 4	0.98 / 0.975 / 0.984 ($\approx$ 1, dial-invariant)
Both-channel	$c = 0.5$ / 0.6 / 0.7 / 1.0	1.99 / 1.86 / 2.15 / 1.89 ($\approx$ 2, dial-invariant)
m-only, clean v	$s = 0.02$ / 0.05 / 0.1 / 0.25	251 / 2,618 / 7,773 / 1.0×10^7 ($\geq$ 250, exploding)

Table 3: Routing pins realized power. Within each routing Γ is nearly dial-invariant, so the injected magnitude is an x-axis the model does not feel near small-gradient states. Corollary (method): gradient-space power matching is not an adequate control; report realized update-space power.

Structural disclosure (finding, not caveat). The v-only routing, with $\Gamma_v \approx 1$, is structurally a rescaled plain-gradient (momentum) process: the noise floor in v flattens Adam's per-coordinate $1/\sqrt{v}$ normalization, so v-only and both-channel are different effective optimizers rather than two settings of one. We report this as a finding—the noise floor in v destroys Adam's normalization—directly analogous to the mechanism by which DP-Adam collapses to DP-SGD absent bias correction [Tang et al., 2024, Li et al., 2022], and to the sign/variance decomposition of Adam updates [Balles and Hennig, 2018, Kunstner et al., 2023]. It is disclosed here, where the structural facts live, rather than treated as a confound to be matched away.

C Corrections ledger (audit trail)

Each row is a claim an earlier draft made and the measurement that retired it. The list is the audit trail for the trust spine; it is reported in full so the retractions are legible.

Retraction	What was claimed	What retired it
Irrecoverability	the collapsed state is a terminal minimum the optimizer would not exit	clean full-gradient replay re-solves in $\sim$800 steps from the identical (θ, m, v); all non-escapes reframed as $\geq$-horizon censoring (Fig. 1)
Erasure-to-scratch	noise resets the solution back to initialization	collapsed+v-noise reacquires at 4,600 vs fresh+v-noise 8,400 on the same table (Fig. 4); the head start survives, so the state is not reset
"v-channel explains everything"	the second-moment channel alone accounts for the trap	single-channel isolation arms cannot be matched on gradient-space power—Adam couples the channels through the shared update; the clean statement is the pinning corollary (App. B)
Crawl / round-trip overcount	a progress metric that double-counted a crawl and its return	the overcount was located and the metric corrected downward
Ignition	an "ignition" trigger initiates escape	the mechanism's own pre-registered test did not fire; demoted
Corridor / reservoir	escape proceeds through a fixed "corridor" / an erasable progress reservoir	progress survives dosing and has no single module address; demoted
Registered horizon deviation	stuck-state detector threshold $T =$ 3,500	reduced to $T =$ 2,000 mid-program; logged and applied uniformly across all arms
Device sensitivity	an absolute escape-time claim	the metastable escape is float/device-sensitive (CPU vs MPS); only the relative across-arm comparison under one harness is used

Table 4: Corrections ledger. The collapsed state is an attractor in the sense of Chen et al. [2023], but recovery is a censored $\geq$-horizon event, not a fixed endpoint.

D Exploratory: the $\Gamma \approx 2$ anomaly

This section is *exploratory* and not load-bearing; the main text reports $\Gamma \approx 2$ only as a measurement. The per-step power admits an exact algebraic decomposition,

$$\Gamma \;=\; 1 + q^2 - 2q\rho, \qquad q = \frac{\|u_{\text{both}}\|}{\|u_{\text{clean}}\|}, \quad \rho = \cos\!\big(u_{\text{both}}, u_{\text{clean}}\big).$$

From the logs, $q = 0.70$ (the both-channel update is *smaller* than clean) and $\rho = -0.28$ (anti-correlated), giving $\Gamma = 1.894$, matching the measured 1.89; the residual is large ($\nu = \|r\|/\|u_{\text{clean}}\| = 1.38$) and anti-aligned ($A_{\text{align}} = \cos(r, u_{\text{clean}}) = -0.87$). The scalar prediction $1 + a^2 = 1.006$ ($a = 0.079$) is thus off by a factor of two.

Sparsity account. The clean update is sparse: its participation ratio is 2,576, i.e. 0.32% of $D =$ 807,720 coordinates. On the remaining $\approx 99.68\%$ near-zero coordinates the per-coordinate $1/\sqrt{v}$ normalization renders the injected floor into a unit-scale sign field, so the both-channel update carries most of its mass in the null directions—this is what drives $\rho < 0$ and the large ν. A β_1 scaling for the momentum channel: the stationary fraction of injected noise surviving the low-pass is

$$f(\beta_1) = \frac{1-\beta_1}{1+\beta_1} \qquad (\, f = 0.0526 \text{ at } \beta_1 = 0.9\,),$$

which, combined with the null-coordinate mass, gives $q < 1$; when the null mass balances the surviving active drift, $\Gamma = 1 + q^2 - 2q\rho \to 2$. The account rests on stated independence assumptions: (i) injected coordinate noise is zero-mean, iid across steps and coordinates, and independent of the clean gradient; (ii) the second moment v is quasi-stationary over the window (frozen denominator); (iii) on null coordinates the clean gradient is negligible, so the normalized update is pure sign-of-noise. Under (i)–(iii) the anisotropy resembles the SGD covariance structure of Zhu et al. [2019], Wen et al. [2020], Ziyin et al. [2024]. We do not fit these; the exact decomposition above is what this section rests on. *Scope note (consistency check, not a discovery):* at the three probed states the escape-direction drift is matched—two decimals at the fork ($\mu_{\|} = 0.498$ vs 0.502, Δ resolved at $\sim 0.8\%$) and exact along the trajectory ($0.375/0.375$; $0.223/0.223$)—confirming that nonlinear (Jensen) corrections through the Adam denominator are negligible *at these matched states*; the load-bearing empirics remain the fates (re-solve vs. $\geq$-horizon censoring), the diffusion ratio, and that these hold in a real curved landscape rather than a constructed one (Fig. 3).

E Dose–response

Table 5 is the v-only inset of Fig. 4: onset step from the collapsed state versus injected scale s, at $\eta = 0.0125$, single seed per cell. Onset rises then saturates; increasing s past 0.75 does not further delay reacquisition, consistent with the pinning result that realized felt-noise, not injected amplitude, sets the trap. This is the opposite of the saddle picture, where added isotropic noise accelerates escape [Jin et al., 2017, Zhu et al., 2019]; the operative regime is set by the realized noise magnitude relative to the stability edge [Wu et al., 2018, Andreyev and Beneventano, 2024].

Injected scale s	0.25	0.50	0.75	1.00
v-only onset (steps)	3,400	3,600	4,600	4,600

Table 5: Dose–response (v-only, collapsed start). Onset saturates in the injected scale.

F Secondary note: entrenchment

A contingent observation, offered only as a secondary note and not used elsewhere. Under v-only noise on the original table, the young collapsed checkpoint reacquires faster than a fresh initialization (4,600 vs 8,400): a recently collapsed state retains a usable head start. Aged collapsed checkpoints reacquire more slowly, consistent with loss-of-plasticity phenomenology [Dohare et al., 2024] and late-stage grokking collapse [Prakash and Martin, 2026]. The ordering is contingent on checkpoint age and table identity, so we do not build on it. The same 2×2 numbers also admit a common-onset-floor reading—noise imposing a shared reacquisition floor largely independent of start state—and we give it equal weight; the entrenchment ordering holds only under the targeted-erasure reading. This bears on the call to study training dynamics as objects in their own right [Biderman et al., 2026].